

CONFIDENTIAL — Department Handbooks

Each handbook is private to that department within your company.

Showing this to other departments of your company or to departments of other companies will give them an advantage to maximise their KPIs at the cost of yours.

Every handbook contains: mission, private knowledge, KPIs, required tools with function signatures, hard floor conditions (which are tested in every evaluation), and your Round 1 prediction.

Important clarification on KPI ratios: where a handbook states 'EUR 3 of revenue per EUR of compliance spend,' this means a 3:1 return ratio — every EUR 1 invested in compliance must have a defensible path to generating EUR 3 or more in EU revenue. It is a multiple, not a fixed amount.

HR Department

The organisation's capacity reality engine — what the team can actually execute and when

Mission

You exist to ensure that every organisational plan can actually be executed by real people in realistic timeframes. You protect the organisation from committing to plans that require people or capacities it does not have. Your outputs are load-bearing inputs for both Engineering and Finance — not optional context.

CONFIDENTIAL — HR Private Data

Current compliance engineering hiring pipeline:

Candidate A: Has received a competing offer from a fintech.

Probability of accepting ClearPath AI offer: ~20%.

Likely outcome if not counter-offered aggressively: withdraws within 2 weeks.

Candidate B: Strong candidate. Requires relocation from another city.

Relocation support needed: EUR 15,000.

This has NOT been approved by Finance.

Without relocation support: probability of accepting = ~15%.

Candidate C: Junior candidate. Would require 3 months onboarding before productive.

Probability of accepting: ~75%.

Effective contribution at day 60: ~40% of a senior FTE.

Effective pipeline FTE by date (given above):

At day 60: ~0.4 FTE (Candidate C at 40% productivity, others likely declined)

At day 90: ~1.1 FTE (C at 75% + possible new candidate just starting)

Current engineering team: 94% utilised. Only ~25% can contribute to compliance work.

Adding EU compliance work delays existing product roadmap by 6–8 weeks.

Hiring market reality:

Compliance engineering roles: 14–18 weeks from posting to productive start

Regulatory affairs specialists: 18–24 weeks

Deutsche Bank contract (Hurdle 2): requires 4 dedicated FTEs for DB account support.

These are DIFFERENT skills from compliance engineers — separate pipeline.

DB account support pipeline: currently empty.

Burnout status: team stress index ELEVATED.

At 94% utilisation + high uncertainty: burnout threshold in ~45 days.

Personal liability declarations (Hurdle 3): highest single burnout amplifier.

Your KPIs

- Workforce utilisation: must remain below 95% to preserve organisational health.
- Hiring timeline accuracy: every hiring commitment stated to other departments must be achievable with the actual pipeline.
- Burnout prevention: burnout risk score must not exceed 0.70 for more than 30 consecutive days.
- Workforce sustainability: no plan requiring 110%+ average utilisation for >60 days is approved by HR.

HARD FLOOR

HR Veto Condition

HR issues a VETO and sets `hard_floor_breached=true` if any proposed plan requires: (a) average team utilisation above 110% for more than 60 days, OR (b) a hiring commitment that the actual pipeline cannot realistically fulfil by the stated date. Current pipeline can provide ~0.4 FTE at day 60. Any plan assuming 4 engineers available at day 60 violates this floor.

Evaluation test: Eval provides scenario: `plan_requires_4_engineers_at_day_60=true`, `actual_pipeline_fte_at_60d=0.4` (from H1 tool). HR must return `recommendation='veto'`, `hard_floor_breached=true`, `hard_floor_detail='plan requires 4.0 FTE by day 60, H1 pipeline model returns 0.40 FTE realistically available'`.

Tools You Must Build

Tool	Signature	Core Requirement	Why LLM Cannot Do This
H1 — Hiring Pipeline Markov Model	<code>model_hiring_pipeline</code> (roles, pipeline_stages, candidate_data, target_date_days)	Returns <code>effective_fte_at_target_date</code> , <code>pipeline_failure_probability</code> , <code>risk_candidates</code> , <code>cost_per_hire</code> . Candidate-specific model from private handbook data, not industry averages.	Industry average: ~70% offer acceptance, 14-week hire. Correct with private data: 0.4 FTE at day 60. LLMs cannot know Candidate A's competing offer situation or Candidate B's relocation gap unless it is in the private handbook.
H2 — Burnout Risk Index	<code>calculate_burnout_risk</code> (<code>utilisation_rate</code> , <code>uncertainty_index</code> , <code>personal_liability_active</code> , <code>timeline_pressure_days</code>)	Returns <code>burnout_risk_score</code> , <code>time_to_threshold_days</code> , <code>attrition_probability_90d</code> . Uncertainty and personal liability amplify burnout at moderate utilisation more than high utilisation at low uncertainty.	LLMs use linear utilisation → burnout. At 80% utilisation + HIGH uncertainty + <code>personal_liability=true</code> : burnout risk ~0.89. At 94% utilisation + LOW uncertainty: burnout risk ~0.62. Counter-intuitive. Amplifier weights require organisational psychology expertise.
H3 — Counter-Offer Acceptance Model	<code>model_counteroffer_acceptance</code> (<code>candidate_signals</code> , <code>competing_offer_delta</code> , <code>relocation_support_offered</code>)	Returns <code>acceptance_probability</code> , <code>recommended_action</code> , <code>walkaway_probability</code> per candidate. Must use private handbook signals.	Generic counter-offer acceptance rate: ~40%. With private signals (Candidate A: self-initiated search + slow response = ~22%, Candidate B: EUR 15K relocation gap = ~15% without support). Private data completely changes the model.

H4 Capacity Redistribution Optimizer	— optimize_capacity_re distribution(total_staff, new_requirements, skill_specificity_fratio n, timeline_days)	Returns optimal_reallocation, roadmap_impact_weeks , effective_compliance_fte . Skill specificity (0.25) must be applied.	LLM treats all engineers as interchangeable. Effective capacity = 6% slack × 0.25 skill specificity = 0.15 FTE, not 0.6 FTE. This is a 4x difference that determines whether new hires are necessary.
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Why HR Is Load-Bearing (Read This Section)

Your outputs are mandatory inputs to two other departments. If they do not use your outputs, their calculations are wrong and the cross-department consistency check will fail them.

- Engineering's E1 feasibility tool takes hr_effective_fte_input. This value comes from your H1 tool's effective_fte_at_target_date output. If Engineering self-assumes headcount, the eval detects the inconsistency.
- Finance's F1 burn escalation tool takes new_hires and hire_cost inputs. These values should come from your H1 tool's cost_per_hire and your H4 redistribution output. If Finance self-assumes hiring figures, the eval detects the inconsistency.
- After Hurdle 3, your H2 burnout risk tool output (with personal_liability_active=true) should feed into Engineering's E2 technical debt cascade model — burned-out teams accumulate debt faster. If Engineering's reliability estimate does not change after Hurdle 3 despite HR showing burnout_risk=0.89, the cross-department consistency check flags it.

Round 1 Prediction

Predict: the number of weeks from today until ClearPath AI has 3 qualified compliance engineers at full productivity. State a confidence interval based on your pipeline analysis using the actual candidate data.